

Deadlines You Can Docket Against

Verifying Computed Trademark Deadlines Against 300,000 Register Records

Signal Research

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Abstract

Most trademark data errors cost analysis quality. Deadline errors cost the trademark. A missed renewal or maintenance filing does not degrade a registration; it cancels it. The calendars that prevent this (docketing systems) are traditionally maintained by hand, office by office, at considerable expense. Signal computes these deadlines: renewal cycles, declarations of use, grace periods, restoration windows, and opposition periods, from statutory rules modeled for 22 jurisdictions and exposed directly in the API.

This report does two things. First, it describes the engine and the standard we hold it to: deadline computation at Signal is a pure function (no database, no network, no clock dependence beyond an explicit as-of date), every rule carries citations to its statutory sources with a last-verified date, and the full rule corpus is itself published through the API for inspection. Second, and mainly, it reports a register-based verification. We froze a pre-registered, hash-manifested sample of 31,547 registrations across 10 production offices, then an expanded same-seed sample (stability check) of 306,896 that is a strict superset of the first (the same seed, so every original record is retained), and measured our computed deadlines against three complementary forms of ground truth: dates the offices themselves publish on the record (for one office, USPTO, the compared date is connector-derived; that cell is flagged and its independent evidence is the event arm), the dated renewal and maintenance filings that actually occurred in register histories, and 34,132 observed opposition proceedings. The rules were corrected against these same records during the evaluation, so the expanded sample is a stability check, not an independent holdout.

The engine does not guess. Where a record does not carry the input the statute requires (three documented cohorts, section 3.4) and the office has stated no expiry to anchor on, it declares that the office's date is required instead of computing a date that the register showed to be wrong a material share of the time. This policy is specified in the product code and applied identically in the evaluation, so nothing below depends on an after-the-fact choice of population.

On the committed 31,547-record sample, every comparable record is either verified or declared: 99.69% (95% CI 99.62 to 99.75) of 29,102 comparable records are either computed and exactly equal to the office-published date, or explicitly declared as requiring the office's date. On the records the engine computes (26,607, or 91.4% of the comparable population), exact agreement between the pure statutory computation, with the office's date withheld, and the office-published date is 99.66% (95% CI 99.58 to 99.72); weighted by each stratum's true production share it is 99.82%. The schedule the API actually serves, which anchors on the office's stated date where one

exists, agrees at 99.65% equal-quota and 99.89% production-weighted; because that path uses the office's date as an input it is a description of what customers receive, not independent evidence of rule correctness. 94.7% of 24,091 observed renewal and maintenance filings on computed records fall inside our computed windows (99.4% at USPTO), 93.2% of all 27,562 on the served schedule, and 90.2% of 23,751 verifiable opposition filings fall inside our computed opposition windows (98.2% at EUIPO, the cleanest cell). Every disagreement was individually classified by mechanism; 90 records disagree after the declared cohorts are set aside, and the genuinely unexplained residual remains 3 records. The v0.91 full-population figure of 96.08% at expanded scale, which scored the engine's guesses on the declared cohorts against it, is retained in section 5.1 for continuity.

The verification also found defects: seven in our own rules (six in the maintenance rules, one in the opposition rules), all fixed, with agreement reported before and after each fix; three in our data ingestion (two fixed, one filed); and 311 records classified by Signa as register-data errors, with per-record evidence, where the register rather than the computation carries the wrong date. A subsequent mutation-testing hardening pass over the rules package (version 1.0, section 3.6) found and fixed a further ten engine defects, none of which moved a verified date on the ten offices. We publish the methodology in full, including the discrepancy taxonomy and the cases we got wrong, because a deadline product asks for a specific kind of trust, and we believe that trust should be checkable.

About this report. This is a first-party evaluation: Signa selected the metrics, wrote the evaluator, corrected the system under test, and classified the disagreements. It has not been independently audited. The artifacts are published so that any reader can check the work.

1. Background: the docketing problem

Registered trademarks are not granted once; they are kept alive. Nearly every jurisdiction requires periodic renewal, and several require affirmative proof that the mark is actually in use: the United States demands a declaration of use between the fifth and sixth year after registration (Section 8 of the Lanham Act), again at each renewal (Section 8 combined with Section 9), and under a separate provision for Madrid-route registrations (Section 71). Miss the window, and after a short grace period the registration is cancelled. Oppositions run on even tighter calendars: a third party typically has a fixed window after publication to oppose, and a brand owner watching for conflicts has exactly that window to act.

What makes this hard to automate is not any single rule but the accumulation of jurisdictional texture:

- **Different triggers.** Renewal terms run from filing in some jurisdictions and from registration in others. Sweden switched triggers on 2019-01-01, and the correct trigger for a Swedish mark depends on which side of that reform it was granted.
- **Route matters.** The same US registration is governed by Section 8 or Section 71 depending on whether it arrived directly or as a Madrid designation, and a Madrid designation's renewal anywhere runs on the international registration's ten-year cycle, not the national trigger.

- **Legacy regimes.** Canadian registrations renewed before the 2019-06-17 reform carry 15-year terms and can sit on mixed 15-and-10-year ladders. Australian marks accepted before 2013-04-15 had a three-month opposition window; later ones have two.
- **Calendar arithmetic is legally defined.** What "three months after" means at a month boundary is specified by statute and differs by country (German law, the Canadian Interpretation Act, the Singapore Interpretation Act, and EU delegated regulation all define it, not identically). Leap days, month-end clamping, end-of-month rules for older German registrations, and weekend or holiday rollover (37 CFR 2.196 in the US, Singapore's excluded-day practice including the eves of three holidays) all change real dates.
- **A deadline is a legal day, not a timestamp.** An opposition window closes at the end of the office's legal day in the office's time zone. A system that treats it as a UTC timestamp will tell a California user that a European deadline closed hours before it legally did.

Law firms and corporate IP departments solve this with commercial docketing systems and manual verification. Developers building trademark features have had no API-native option: general trademark data APIs return raw dates from the register and leave the statutory arithmetic to the caller.

1.1 Why verification, not just testing

The engine described here is extensively unit-tested against hand-derived golden dates (section 3.6). But hand-derived goldens share a failure mode with the rules they test: both are produced by humans reading statutes. The strongest available check is the register itself: offices publish expiry and renewal-due dates on records, register histories contain the dated renewal and maintenance filings that actually happened, and opposition proceedings carry the dates oppositions were actually filed. If our computed deadline for a real registration disagrees with what the office printed on that registration, one of us is wrong, and either answer to that question is worth publishing. That comparison, at scale, across offices, is this report.

2. What is computed

The engine produces two families of output.

Maintenance deadlines. For a given mark (jurisdiction, filing route, status, and the record's date fields), the engine returns the schedule of lifecycle deadlines: renewal (recurring by statutory cycle, including legacy-term ladders where a reform changed the cycle length), declaration of use (US Sections 8 and 71), combined renewal and use filings (US Section 8 with Section 9), declaration of incontestability (US Section 15, the one optional entry), restoration windows following a missed deadline, and international-

registration renewal for Madrid records. Each computed deadline carries the date the window opens, the due date, the grace-period expiry where one exists, a status relative to an explicit as-of date (future, window open, due soon, in grace, missed), an urgency tier, and the consequence if missed (cancellation, expiration, or lapse of the international registration).

Opposition windows. For a published application or registration, the engine computes the statutory opposition window: trigger event (publication for opposition, registration publication, or international-designation publication, depending on the office), open and close dates, the routinely available extension where office practice has one, and a must-act-by moment expressed as the end of the office's legal day in the office's time zone with daylight saving handled. Each computed window carries the citation and version of the rule that produced it.

3. The engine

3.1 Rules as audited data, not buried code

Rules are data: 23 jurisdiction configurations covering 22 jurisdictions for maintenance deadlines (the United States carries two, one for direct registrations and one for Madrid designations), and 27 opposition-window rules covering both direct and Madrid-designation routes at the covered offices. Every configuration carries citations to its statutory and office sources, each with the date it was last checked, and a last-verified date for the configuration as a whole. Computed opposition windows embed the rule version that produced them, so any alert or API response can be traced to the exact rule text it came from.

The rule corpus is not a secret. It is published, with citations, through the API (GET /v1/deadline-rules and GET /v1/opposition-rules), so a customer or their counsel can audit the rules we apply before relying on them. The full deadline rule corpus is published for inspection.

3.2 Purity as a correctness strategy

Deadline computation is a pure function: no database access, no network calls, and no hidden dependence on the current clock (callers pass an explicit as-of date; statuses and urgencies are computed relative to it). Identical inputs always produce identical output. This is a correctness strategy, not an implementation detail: pure functions can be property-tested exhaustively, reproduced exactly in an investigation, and recomputed identically at read time, so there is no stored deadline to go stale when a rule is corrected.

Dates are treated as calendar dates, not timestamps, except at the one place time zones legally matter: the must-act-by moment of an opposition window, which is computed as the end of the office's legal day in the office's own time zone. Date arithmetic follows the

applicable legal definitions: month addition clamps to month-end where statutes say so, leap days resolve the way the relevant interpretation acts specify, older German registrations get the end-of-month rule, pre-reform Canadian registrations get the legacy-term ladder, the Swedish trigger change applies from its statutory date keyed on the registration date, and deadlines roll forward over weekends and office holidays against 24 office holiday calendars maintained for 2024 through 2030 (including, for Singapore, the office's excluded-day eves).

3.3 The office's own dates as an anchor

Two behaviors connect the statutory computation to what the office has already said about a record. First, when the office states an expiry date and the statutory ladder computed from the record's trigger date passes through it exactly, the office has confirmed the computation and the full trigger-derived schedule is used. When the stated expiry does not sit on the computable ladder (because a discriminating date is absent from the record, as with Madrid subsequent designations, or because a legacy cadence predates the modeled regime), the schedule is re-anchored on the office's stated date. Second, a renewal cycle that fell due before the office-stated expiry has by definition already been paid (that is why the term now runs to the stated expiry), so it is not shown as an actionable deadline. Both behaviors are measured separately in section 5: the pure statutory computation is evaluated with the office's dates withheld, precisely so that the evaluation cannot be circular.

3.4 Declared computability (version 1.0)

Some records do not carry the input the statute needs, and no amount of arithmetic recovers it. Three cohorts are documented: Australian direct marks filed before 1996-01-01, where the 1955 Act's seven-and-fourteen-year cadence versus the 1995 Act's ten-year term turns on a registration date the register records as equal to the filing date, so roughly one in six of the cohort is on the modern ladder and indistinguishable from the rest; Singapore direct marks filed before 1999-01-15, on the repealed Cap. 332 cadence, with the same problem; and Madrid designations whose row carries no international-registration date, where the renewal term runs on the IR's anniversary (Madrid Protocol Art. 6(1) and 7(1)). For the third cohort the engine first looks up the WIPO parent registration we ingest (by IR number) and anchors on its date; this is exact for original and subsequent designations alike. Where no parent is available, a designation's own filing or registration date is used only at offices where this evaluation verified that it equals the IR date (near-100% agreement per office: cipo, nipo, ipau, ipi and wipo on the filing date; euipo and prv on the registration date). Elsewhere, notably IPOS, whose priority-linked international marks carry neither column as the anchor, no stand-in is used.

When the office states an expiry date, the anchored schedule of section 3.3 is exact and the record is computed. Otherwise the engine declines: it returns no schedule and reports `requires_office_date` with the cohort, on `POST /v1/deadlines/compute` as supported: `false` with that reason. This replaced the previous behavior of computing a guess for these records, and it replaced the previous evaluation's post-hoc "modeled population": the same rule now lives in the product and in the harness, so the population the engine is scored on is the population it claims to compute. A declared record is scored as correct behavior (no wrong date was served) and reported separately; it is never counted as agreement.

3.5 Exposure and cost

Computed deadlines appear on the trademark detail endpoint (`GET /v1/trademarks/{id}`), on portfolio deadline views (including an iCalendar feed), and inside watch alerts, where the opposition must-act-by is computed with the provenance fields attached. For bulk and integration use, two dedicated endpoints compute without persisting anything: `POST /v1/deadlines/compute` and `POST /v1/oppositions/compute` accept up to 1,000 items per call, take an explicit as-of date and horizon, and return per-item results including an explicit supported: `false` with a reason for jurisdictions we have not modeled, rather than a silent guess. The engine computes full schedules for the entire 31,547-record evaluation sample in roughly 256 milliseconds single-threaded on a laptop, about 8 microseconds per record.

3.6 Standing tests and the frozen regression gate

The rules package ships with roughly 1,450 test cases: 31 plain-language specification files (one per jurisdiction plus the engine mechanics, each scenario an executable test), per-office golden dates for maintenance and opposition computations, golden pins over every statutory configuration, calendar-arithmetic and holiday-rollover suites, and property-based tests executed across every jurisdiction configuration. In August 2026 the package was put through mutation testing: 2,946 mutants across all 57 files, every one killed by a test or classified in place with a reason (100%); the same was done for the compute endpoint. That pass found ten engine defects, all fixed and re-verified against the frozen sample with no movement on the ten verified offices; they are listed in the changelog.

A frozen regression gate runs in continuous integration on every change: it recomputes the 31,547-record maintenance sample and pins that sample's Arm A agreement and declared counts, its Arm B totals, and every Arm C cell of the opposition sample at the frozen as-of date, so an engine change that moves a single record fails the build until it is re-verified cell by cell. The gate does not cover the expanded 306,896-record run or the production-weighted outputs.

4. Verification methodology

4.1 Design principles

Four commitments govern the design. The sample was frozen (record identifier lists hashed with SHA-256) before any comparison ran. Every number carries its n , headline numbers carry 95% confidence intervals (Wilson), and no cell below $n=50$ is reported. Every disagreement is individually classified by mechanism, not left in an "other" bucket. And when the evaluation itself exposed a defect in a rule, we corrected the rule and report agreement both before and after the correction. No correction was accepted on empirical fit alone; each required a statutory or documented-data-source justification, verified independently against primary sources, and was required to fit the register data without regressing any other cell. Several implementation choices (anchor selection, era boundaries) were informed by the data and are disclosed as such. Correcting rules against the evaluation and then reporting only the corrected number would make the exercise circular; showing both makes a correction evidence of the process working. One protocol amendment is disclosed rather than left silent: the pre-registered comparison horizon of 15 years was raised to 45 years during the evaluation after it produced false disagreements on old Canadian records (section 5.7); a wider horizon cannot create false matches.

4.2 Samples

Three frozen datasets are used.

- **Maintenance sample.** A deterministic, seeded sample of 31,547 registrations across 10 production offices (USPTO, EUIPO, WIPO, CIPO, IP Australia, IPOS, the Norwegian, Swedish, and Swiss offices, and INPI France), stratified by filing route (direct versus Madrid designation) and registration era at 500 records per stratum, to exercise legacy-regime rules deliberately.
- **Expanded same-seed sample (stability check).** The same design at tenfold depth: 306,896 records, drawn with the same seed, yielding 282,393 comparable records. Because the seed is identical, this is a strict same-seed superset of the development sample (every original record is retained), not an independent holdout: the rules were corrected against these same records. It checks whether the headline rates are numerically stable at greater depth, not whether the corrected system generalizes to unseen records.
- **Opposition sample.** 34,132 opposition proceedings with observed filing dates across the five offices whose proceedings data carries oppositions (USPTO, EUIPO, CIPO, IP Australia, IPOS), hash-ordered per office and route cell.

Records with statuses under which no deadline is defined were excluded by pre-registered rule. Because equal-quota strata overweight disagreement-heavy populations (legacy eras, Madrid designations at specific offices) relative to production, the headline is also reported reweighted by each stratum's true production population (9,166,622 records). The weighted Arm A figures and the pooled Arm B and Arm C figures do not yet carry design-based uncertainty intervals; a simple binomial interval would understate the uncertainty because events and comparisons cluster within records and strata, so those intervals are queued as future work.

4.3 Three arms of ground truth

- **Arm A, office-published dates.** For offices that state an expiry or renewal-due date on the record, the computed next renewal deadline must equal the office's stated date exactly. Disagreements are additionally banded by distance. Weekend and holiday filing rollover is treated as a filing-day convention, not an expiry disagreement. The office's stated expiry is withheld from the engine input in this arm, so the computation being tested cannot see the answer (section 3.3). Under the declared- computability policy (section 3.4) a record the engine declines is reported as declared, counted as correct behavior in the verified-or-declared figure, and excluded from the agreement denominator; it is never counted as agreement.
- **Arm B, event consistency.** For offices whose register histories carry dated renewal and maintenance filings, every observed filing must fall inside the window we computed for it (window open through grace expiry). USPTO events are restricted to a pre-registered allowlist of filing-coded prosecution events, excluding courtesy reminders and office notices, which measure office latency rather than rule correctness. Events on declared records are not scored in the pure arm (the engine claims no window for them) and are counted; the served-schedule figure scores all events.
- **Arm C, observed oppositions.** Every observed opposition must have been filed inside the computed opposition window for the opposed record's publication date, counting the rule's modeled routine extension separately. One non-circular validity filter applies: proceedings data attaches an opposition to both parties' records, and a record already registered when the opposition was filed cannot be the opposed application, so those rows are excluded by registration timing alone.

Two circularity guards matter for reading Arm A. USPTO publishes no expiry date (Signa derives one), so only actually-renewed USPTO records enter Arm A and the USPTO cell is the weakest in the table by construction; the USPTO evidence that matters is Arm B. And where an office publishes a renewal-due date it equals the expiry date in every feed except WIPO, so it is not an independent signal and Arm A uses the stated expiry only.

5. Results

All numbers are reproducible from the committed frozen data and results artifacts; runs are identified by date in the changelog of the evaluation record.

5.1 Headline

All figures below are from the committed 31,547-record sample re-run on 2026-08-30 under the version 1.0 policy (section 3.4), except where marked v0.91.

Metric	Value	95% CI	n	Notes
Schedule the API serves, agreement with office-published dates, production-weighted	99.89%		estimated over 9,166,622 production records from n=29,102	anchors on the office's stated date where one exists; describes what customers receive
Schedule the API serves, equal-quota	99.65%	[99.57, 99.71]	29,102	
Verified or declared (exact agreement + declared, over all comparable records)	99.69%	[99.62, 99.75]	29,102	primary version 1.0 figure
Pure statutory computation, office's date withheld, computed records	99.66%	[99.58, 99.72]	26,607	engine correctness on the population it claims
Pure statutory computation, production-weighted	99.82%		estimated over 8,965,731 production records	
Records declared (requires the office's date)	2,495		8.57% of 29,102	AU direct pre-1996: 363; IPOS direct pre-1999: 592; IPOS Madrid designations: 1,540
Arm B event consistency, computed records	94.71%	[94.42, 94.99]	24,091 events	3,471 events on declared records not scored
Arm B event consistency, served schedule	93.18%		27,562 events	
Arm C opposition consistency	90.23%		23,751 verifiable oppositions	unchanged; now CI-pinned per cell

Metric	Value	95% CI	n	Notes
Disagreements remaining after the declared cohorts	90		0.34% of computed	all classified, section 5.5
Genuinely unexplained residual	3 records			
vo.91 full-population exact agreement (engine guesses scored against it)	96.08%	[96.00, 96.15]	282,393 (expanded same-seed sample)	retained for continuity; 96.00% on the committed sample
vo.91 modeled population (post hoc)	99.74%	[99.72, 99.76]	282,393	superseded by the declared policy, which specifies the same three cohorts in code

The primary version 1.0 result is the verified-or-declared figure: every comparable record is either computed and exactly equal to the office's published date, or explicitly declared as requiring that date. The engine correctness figure isolates the arithmetic: on the records the engine claims to compute, with the office's date withheld, it is 99.66% exact. The served figures describe what a docketing customer experiences and are partially self-referential. The vo.91 full-population figure is retained so the change of framing is visible: it scored the engine's guesses on the declared cohorts, which the engine no longer makes. The expanded 306,896- record stability check of vo.91 is not re-run here; its purpose (rate stability at depth) was met and the declared policy removes the cohorts that drove its difference from the committed sample.

5.2 Arm A by office and route (pure computation, office's date withheld, version 1.0)

Committed sample; exact agreement with the office-stated expiry on computed records, with the declared count per cell.

Office	Route	n computed	Exact agreement	Declared
EUIPO	direct	2,000	100.00%	0
EUIPO	madrid	2,000	99.90%	0
WIPO	madrid (IR)	1,049	100.00%	0
Switzerland (IPI)	direct	2,000	99.90%	0

Office	Route	n computed	Exact agreement	Declared
Switzerland (IPI)	madrid	2,000	99.80%	0
Sweden (PRV)	direct	1,505	99.93%	0
Sweden (PRV)	madrid	1,500	100.00%	0
Norway (NIPO)	direct	2,000	99.35%	0
Norway (NIPO)	madrid	2,000	100.00%	0
Canada (CIPO)	direct	2,000	99.65%	0
Canada (CIPO)	madrid	500	100.00%	0
France (INPI)	direct	2,000	98.40%	0
USPTO	direct (derived expiry)	1,502	99.80%	0
Australia (IPAU)	direct	1,637	99.94%	363 (filed before 1996-01-01)
Australia (IPAU)	madrid	1,500	99.80%	0
Singapore (IPOS)	direct	1,408	98.44%	592 (filed before 1999-01-15)
Singapore (IPOS)	madrid	0	(declared)	1,540

Three cells moved materially against vo.91, all by mechanism: Australia direct (84.90% to 99.94%) and Singapore direct (82.75% to 98.44%) because the legacy-cadence cohorts are now declared instead of guessed; and the Madrid cells at Canada (90.60% to 100.00%), Australia (89.20% to 99.80%), Norway and Sweden because designations now anchor on the WIPO parent's international-registration date (209 former disagreements, all resolved exactly). Singapore's Madrid designations are declared in this arm: IPOS is not an office where the designation's own dates were verified as the IR date, and its priority-linked international marks carry no anchor on the row. On the served schedule (office expiry present) those 1,540 records agree at 97.01% and Singapore direct at 97.90%; the residual there is the register-error class of section 5.5.

5.3 Arm B by office and route (events against windows, version 1.0)

Office	Route	Events (computed records)	Pure	Events (all)	As served
USPTO	direct	4,426	99.39%	4,426	99.39%
USPTO	madrid	1,956	99.59%	1,956	99.59%
Australia (IPAU)	direct	3,308	99.67%	6,779	90.50%
Switzerland (IPI)	direct	2,014	99.40%	2,014	99.40%
Australia (IPAU)	madrid	1,830	98.25%	1,830	98.25%
Sweden (PRV)	madrid	448	97.54%	448	97.54%
France (INPI)	direct	2,993	90.11%	2,993	90.61%
Sweden (PRV)	direct	5,006	87.91%	5,006	87.91%
Canada (CIPO)	direct	2,087	86.97%	2,087	87.49%

The USPTO cells are the cleanest test of the most intricate rules in the system (Section 8, 9, and 71 declaration windows including grace): 99.4% of 6,382 independent filing events fall inside the computed windows. The Australian direct cell, 57.62% in v0.91, is 99.67% once the 3,471 events on declared pre-1996 records are set aside (the engine claims no window for them); on the served schedule, which anchors those records on the office's expiry, the same events verify at 90.50%.

The remaining misses cluster by mechanism. Sweden direct: 470 of 605 misses are filings BEFORE the computed window opened (median 123 days early), consistent with the twelve-month renewal window of the pre-2019 Swedish Act, which is not modelled (the same class as Norway's pre-2024 window, section 7). France: 268 of 296 misses are recorded AFTER grace expiry (median 109 days), a register recording lag rather than a window error. Canada: 171 of 272 misses are dated about 9.5 years before the computed window, that is, on a ten-year cadence where the legacy ladder computes fifteen; this residual is filed. A variant metric that also counts an event dated exactly at a computed cycle anniversary changed no cell: the "renewed-until anniversary" recording convention hypothesized in v0.91 does not explain these misses, and that sentence is withdrawn.

5.4 Defects the verification found in our own rules, all fixed

Each fix was verified against primary legal sources and required to fit the register data with no regressions in any other cell. Pre-fix and post-fix agreement is reported for each.

Finding	Statutory basis	Effect of the fix
1. The WIPO configuration was keyed to a different code than production records carry, so every international registration silently received zero computed deadlines	Registry alias, no rule change	766 newly covered IR records agree at 100.00%
2. Canadian legacy renewal cadence: the 2019 reform shortens each term only when the period starts on or after the cutoff, so pre-2019 marks renew on successive 15-year terms first	Trademarks Act (Canada) s.46, in force 2019-06-17	CIPO direct Arm A 73.15% to 99.65%; Arm B 64.73% to 86.97%
3. Madrid designations renew on the international registration's ten-year anniversary, not the national trigger	Madrid Protocol Art. 6(1), 7(1)	CIPO madrid 0.00% to 90.60%; Norway madrid 45.75% to 99.90%; EUIPO madrid 70.30% to 99.90%
4. The German end-of-month rule leaked onto German Madrid designations	End-of-month rule applies to national registrations only	26 records corrected, up to 30 days
5. The Swedish 2019 trigger reform was keyed on the filing date instead of the registration date, mis-anchoring marks filed before but granted after the reform	Swedish reform of 2019-01-01	5 records corrected by five to seven years
6. US Section 71 windows were anchored on the protection-statement date; the statute keys every Section 71 window to the certificate issuance date, the US registration date	15 U.S.C. 1141k(a); 37 CFR 7.36(b); TMEP 1613.04	USPTO madrid Arm B 66.16% to 99.59%

Finding 6 deserves a note on method: an anchor sweep over 1,956 observed Section 71 filings scored the registration date at 99.39% against 66.16% for the previously used field and at most 64% for every other candidate, and the statutory text, pulled independently, names the same date. Data and statute converged before the rule changed.

5.5 Classification of every remaining disagreement

All 544 in-scope structural disagreements at expanded scale were classified mechanically, each against production event and filing history, with every claimed engine defect adversarially re-verified:

Class	Count (v0.91, expanded sample)	Disposition in version 1.0
Classified by Signa as a register-data error, with per-record evidence	311	Still counted against the computed population; the 90 committed-sample disagreements that remain are this class (France renewals recorded as new deposits under the former law, 32; Singapore direct post-1999, 22; the Norwegian bulk-load placeholder expiry, 13; Canada, 7; Switzerland, 6; Australia, 4; USPTO derived, 3; EUIPO, 2; Sweden, 1)
Anchoring date absent from the record (Madrid subsequent designations, Singapore priority-linked international marks)	225	Resolved by mechanism: designations with a WIPO parent now anchor on its IR date (exact); IPOS designations without an anchor are declared; the served schedule anchors on the office's expiry
Records mis-routed by ingestion (Madrid designations ingested as direct)	5	Ingestion fix shipped; classification, not rule, error
Genuinely unexplained	3	Open

Zero claims survived adversarial verification as a true rule-logic defect: every cohort initially flagged as an engine bug reduced to a missing input or a register error. The register-error class includes French renewals recorded as new deposits under the former law, a Norwegian bulk-load placeholder expiry shared by unrelated marks, and ancient Canadian statutory-anchor phase offsets. Registers are large and contain errors; that a verification of this kind surfaces them, with evidence, is the strongest argument for computing deadlines rather than echoing stored ones.

5.6 Opposition windows against 23,751 observed oppositions

Office	Route	Verifiable n	Consistency
Canada (CIPO)	madrid	1,368	99.85%
EUIPO	direct	4,000	98.22%
EUIPO	madrid	4,000	97.95%
Australia (IPAU)	madrid	1,500	94.80%
Canada (CIPO)	direct	4,000	90.83%
Australia (IPAU)	direct	1,839	90.54%

Office	Route	Verifiable n	Consistency
Singapore (IPOS)	madrid	1,446	87.00%
Singapore (IPOS)	direct	3,991	75.90%
USPTO	direct	1,504	75.27%
USPTO	madrid	103	75.73%
Pooled		23,751	90.23%

This arm found two more defects of the silent-mismatch kind, both fixed. The first is a rule defect (the one opposition-rule defect, which brings the rule-defect total to seven: the six maintenance-rule defects of section 5.4 plus this one); the second is an ingestion defect. (7) The IP Australia opposition rule was keyed to an office code production records do not carry, so every Australian record served a null opposition window until this evaluation noticed. (8) The Singapore connector stored the latest gazette republication as the record's first publication date (the office republishes after amendments and registration), which made 749 sampled oppositions appear to have been filed before publication. Every one of them falls inside the two-month window of the earliest gazette entry; the ingestion fix applies to newly synced records, with a backfill of existing records filed. The evaluation also drove three rule additions verified against it: the Australian window's era split (three months before the 2013-04-15 reform, two after, with the routinely recorded one-month structural extension for the two-step filing process), and dedicated Madrid-designation opposition rules at CIPO, IP Australia, and IPOS, of which only Canada's differs from the domestic rule (a single extension hard-capped at four months by Trademarks Regulations s.125). The Canadian Madrid cell then verifies at 99.85%, with 2 of 1,368 oppositions beyond the statutory cap.

The USPTO cells are depressed by a known data property, not the window arithmetic: proceedings data attaches an opposition to both parties' records, the registration-timing filter can only remove the opposer-side rows it can prove, and the residual splits roughly half extension-boundary cases at the 180-day statutory ceiling and half remaining wrong-side linkage (real, decided oppositions attached to a mark whose single publication was years earlier, which no opposition window can contain). The linkage correction is filed as ingestion work; the Singapore direct cell will be re-measured after the publication-date backfill.

6. Analysis

Misses cluster by mechanism, not randomly. Every material disagreement cohort in this evaluation bucketed cleanly by office, route, or era: a legacy cadence, a wrong anchor column, a keying mismatch, a data-linkage property. Nothing material diffused

randomly, which is what one expects if the arithmetic is right and the residual causes are structural. The off-by-N distribution says the same thing: day-level discrepancies are 0.38% of all comparisons; structural misses land months or years off.

The office's date and the statute check each other. The pure computation and the office-stated expiry agree on 99.74% of the modeled population, which means each is a working audit of the other. Where they disagree, adjudication attributed the disagreement to the register more often than to the engine (311 records classified as register-data errors, against no further rule defect identified among the investigated residuals). This is the argument for computing deadlines from statute rather than echoing the register: the computation catches register errors, and the register catches rule errors, and this report is the record of both directions working.

Recomputation at read time beats stored deadlines. Seven rule defects were fixed during this evaluation (six in the maintenance rules, one in the opposition rules). Because deadlines are computed at read time from a pure function, every fix applied to every record instantly, with no re-audit of stored calendars. The before-and-after numbers in section 5.4 were produced by re-running the same frozen sample through the corrected engine, which is exactly what production did the moment each fix deployed.

Arm B is asymmetric, and we score it conservatively. An observed filing inside our window proves the window was usable in practice; the absence of an event proves nothing, and an event recorded outside the window is scored against us whatever its cause. The residual misses have mechanisms (section 5.3): early filings under a pre-reform Swedish window we do not model, late-recorded French renewals, and a Canadian legacy-cadence residual. None is excluded. Version 0.91 attributed part of the residual to a "renewed-until anniversary" recording convention; a metric variant that credits exactly that convention changed no cell, so that explanation is withdrawn.

Declaring is a feature. The largest change in version 1.0 is not a number but a behavior: the engine now states, per record and with a reason, when it needs the office's date instead of computing one. For the three cohorts concerned, the register had shown the computed guess to be wrong often enough that serving it was a liability; a docketing system that says "I need the office's date here" is more useful than one that is confidently wrong one time in six. Because the same rule is specified in the product and applied in the evaluation, the population the engine is scored on is, for the first time, exactly the population it claims to compute.

7. Limitations

- **Coverage is 22 jurisdictions, not all.** Japan, Korea, and China are among the offices whose records Signa carries but whose deadline rules are not yet modeled; the API says supported: false for them rather than guessing, and this report claims nothing about them.

- **8.6% of comparable records are declared, not computed.** The engine states that it needs the office's date for Australian direct marks filed before 1996, Singapore direct marks filed before 1999, and Singapore Madrid designations. In production every one of those offices states an expiry, so the served schedule anchors on it; the declared share is a limit on the pure statutory computation, disclosed rather than hidden. Extending Singapore's Madrid coverage requires the WIPO-parent join to reach its priority-linked international marks.
- **Madrid stand-ins are verified per office, not universal.** Where no WIPO parent is found, a designation's own date is used as the IR date only at the seven offices where this evaluation verified that equivalence; elsewhere the engine declares.
- **The USPTO Arm A cell is derived, not office-stated.** USPTO publishes no expiry date, so that one cell shares derivation logic with the thing it tests. The USPTO evidence that matters is Arm B: 99.4% of 6,382 independent filing events inside computed windows.
- **Four offices have no event ground truth.** EUIPO, IPOS, Norway, and WIPO emit no renewal events in our pipeline, so their results rest on Arm A only.
- **Opposition verification is partial where the data is.** The USPTO opposition cells are bounded by a proceedings-to-record linkage property described in section 5.6, and the Singapore direct cell awaits the publication-date backfill. The EUIPO cells (n=8,000, no extension regime to absorb error) are the cleanest external check of the opposition engine.
- **Maintenance-deadline holiday rollover is opted in by 3 configurations.** Opposition windows apply office holiday calendars across all covered offices; on the maintenance side ten calendars are mapped but only the USPTO (domestic and Madrid) and IPOS configurations opt into rollover, and it is a small number of days at window edges elsewhere. (v0.92 correction: v0.91 said "wired for 10 offices".)
- **Some calendar data is forward-looking.** Rule-computed office holiday calendars run to 2100 (v0.92; previously 2050); the lunar, Islamic and administratively-set calendars (EUIPO, IPOS, IPOPHL, DIP, IP Vietnam, JPO, KIPO, CNIPA) are curated through 2030; a small number of future entries are projected pending official publication and marked as such in the data.
- **One office's renewal trigger is approximated.** The Philippine renewal trigger is computed from registration cycles where the underlying renewal event is not itself carried on the record.
- **The list-endpoint deadline filter is an approximation.** Filtering trademark lists by upcoming deadline uses a stored renewal-due column; the authoritative computation is the detail endpoint and the compute endpoints, which is what this report verifies.
- **Ground truth is the register, and registers contain errors.** That cuts both ways: Arm A disagreements are individually adjudicated rather than assumed to be

our fault, and the register-error class in section 5.5 is evidence-backed per record, not a convenient bucket.

- **Rule verification is dated, not continuous, but regression is gated.** Every rule carries the date it was last verified against sources (the configurations touched by this evaluation carry 2026-07-11 or later). Statutes change; the report version will be bumped when rules or measurements are refreshed. Between refreshes, the frozen-sample regression gate of section 3.5 prevents silent movement.

8. Reproducibility and versioning

- The sample manifests (seeded queries and SHA-256 record-identifier digests) were frozen before comparison ran. The evaluation harness, metrics, frozen samples, per-run results, and the mechanism-level classification of every disagreement are committed artifacts, and the headline counts are pinned in continuous integration against the frozen sample.
- The rule corpus this report verifies is publicly inspectable, with statutory citations, via `GET /v1/deadline-rules` and `GET /v1/opposition-rules`.
- The artifacts are published at <https://github.com/signa-so/research> (folder `deadline-verification`): the frozen manifests, the samples, the events, the oppositions, and the per-record comparison results, so the adjudication of every disagreement can be checked, not taken on faith.
- Source data is public register data; no customer data appears in any evaluation set.
- Published numbers are never edited in place; updated measurements ship as a new version with a changelog entry.

Changelog

- **v1.0 (2026-08-30):** Declared computability. The engine declines, with a stated reason (`requires_office_date`), where a record lacks the statutory input and no office expiry anchors it (AU direct filed before 1996-01-01, SG direct filed before 1999-01-15, Madrid designations with no international-registration anchor); Madrid designations now anchor on the WIPO parent registration's date where one is ingested (209 former disagreements resolved exactly), and designation-date stand-ins are used only at the seven offices where they were verified. The post hoc "modeled population" is replaced by this pre-specified policy, applied identically in the product and the harness. Headline re-framed: verified or declared 99.69% of 29,102; pure engine correctness 99.66% on computed records (99.82% production-weighted); served schedule 99.65%/99.89% (unchanged); Arm B 94.71% on computed records (93.18% served); 90 disagreements remain, all classified; 3 unexplained. Arm B tables and the section 6 Arm B analysis rewritten (the "renewed-until anniversary"

explanation is withdrawn after a direct test). Rule-computed holiday calendars extended to 2100. The vo.91 full-population figure is retained in section 5.1 for continuity.

- **vo.92 (2026-08-30):** Hardening pass, no re-measurement of the headline. A mutation-testing hardening run over the rules package (57 files, 2,946 mutants, 100% killed or classified) and the compute endpoint produced 34 findings and a fix wave. None changes an Arm A or Arm B date on the ten verified offices; one deliberate pin move: the Arm A pure count moved from 27,931/29,096 to 27,937/29,102 because six WIPO-sample rows with jurisdiction MX now compute. Those six carry a filing date before the 2020-11-05 cutoff and no grant date, so once the Mexican 2020 era split is keyed on the filing date (LFPPI Transitorio DÉCIMO, verified against the live IMPI register) they resolve to the pre-LFPPI filing anchor instead of having no resolvable trigger at all. All six agree with the office-stated expiry. Arm C is now evaluated at the frozen as-of date and pinned per cell in CI; the pin reproduces the vo.91 §5.6 cells exactly (pooled 90.23%). Rule corrections outside the verified offices: Danish 2019 era split keyed on the registration date (Lov 1533/2018 §8 stk. 3); Philippine third-year DAU window runs from the filing date (RA 8293 §124.2); Thai opposition 90-day era before 2016-07-28. Engine semantics: the due day and the last grace day now read as actionable (matching the opposition engine); expired/cancelled marks keep their open late-renewal and restoration windows; restoration is derived from missed renewals; the opposition coverage-skip threshold shares the rollover bound (20 days, was 14). §7 corrected: maintenance rollover is opted in by 3 configurations, not "wired for 10 offices".
- **vo.91 (2026-07-12):** Framing corrections after an external methodological review. Full-population agreement (96.08% expanded scale, 96.00% committed small sample) is now the primary headline; the 99.74% figure is labeled a post hoc modeled-population result with its exclusions named, and the 99.89% figure is labeled office-date-anchored and partially self-referential. Removed the "holdout" and "confirmatory" framing (the expanded run is a same-seed superset stability check), narrowed the CI regression-gate claim to what it actually pins, disclosed the 15-to-45-year horizon protocol amendment, reconciled the rule-defect count at seven (six maintenance plus one opposition), softened first-party classification language, added a first-party-evaluation disclosure, and wired the published artifact links.
- **vo.9 (2026-07-12):** All results populated from the completed evaluation: pre-registered 31,547-record sample, 306,896-record expanded same-seed sample, 34,132-opposition sample. Six maintenance-rule fixes, one opposition-rule fix, two ingestion fixes, and three rule additions documented with pre-fix and post-fix numbers. Engine sections re-verified against the shipped codebase (23 maintenance configurations, 27 opposition rules, 24 holiday calendars, approximately 346 rules-package tests).

- **vo.1-draft (2026-07-07):** Internal draft. Engine sections verified against the codebase; results pending the frozen evaluation run.

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